



5.2 Differentiation rules and tangents/normals

Calculus · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Power rule: $\frac{d}{dx}(x^n) = n \cdot x^{(n-1)}$. Chain rule: $\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$. Product rule: $\frac{d}{dx}(uv) = u'v + uv'$. Quotient rule: $\frac{d}{dx}(u/v) = (u'v - uv')/v^2$. At a point $x=a$, the tangent line has gradient $f'(a)$ and equation $y-f(a)=f'(a)(x-a)$; the normal is perpendicular to the tangent, with gradient $-1/f'(a)$, equation $y-f(a) = -1/f'(a) \cdot (x-a)$.

Worked example: follow the reasoning

Find the equation of the tangent to $y=x^3-2x+1$ at $x=2$.
 $y(2)=8-4+1=5$. $dy/dx=3x^2-2$, at $x=2$: $3(4)-2=10$.
Tangent: $y-5=10(x-2) \rightarrow y=10x-15$.

Quick check · multiple choice

For $y=(2x+1)^3$, dy/dx equals:
A $3(2x+1)^2$
B $6(2x+1)^2$
C $6(2x+1)$
D $(2x+1)^2$

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 Consider the curve $y = x^2 \cdot e^x$. [7]
(a) Find dy/dx using the product rule.
(b) Find the gradient of the normal to the curve at $x=1$.
(c) Find the equation of the normal at $x=1$ in the form $y=mx+c$.

Final answer: _____



Answers and explanation

Question 1 [7]

(a) $u=x^2$, $v=e^x$, $u'=2x$, $v'=e^x$. $dy/dx = 2x \cdot e^x + x^2 \cdot e^x = e^x(2x+x^2) = x(2+x)e^x$.

(b) At $x=1$: $dy/dx = 1(2+1)e^1 = 3e$. Gradient of normal $= -1/(3e)$.

(c) $y(1)=1^2e^1=e$. Normal: $y-e = -1/(3e)(x-1) \rightarrow y = -x/(3e) + 1/(3e) + e$.

Quick check answer

B. Chain rule: $dy/dx = 3(2x+1)^2 \times 2 = 6(2x+1)^2$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AA subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.